Topic 1: Amenorrhea (Primary & Secondary Workup)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Absence of menses. Primary: absence of menarche by age 15 with normal secondary sex characteristics, or age 13 without. Secondary: cessation of regular menses for 3 months (or 6 months if irregular). MOST COMMON CAUSE OF SECONDARY IS PREGNANCY.

CLINICAL PRESENTATION & PHYSICAL EXAM

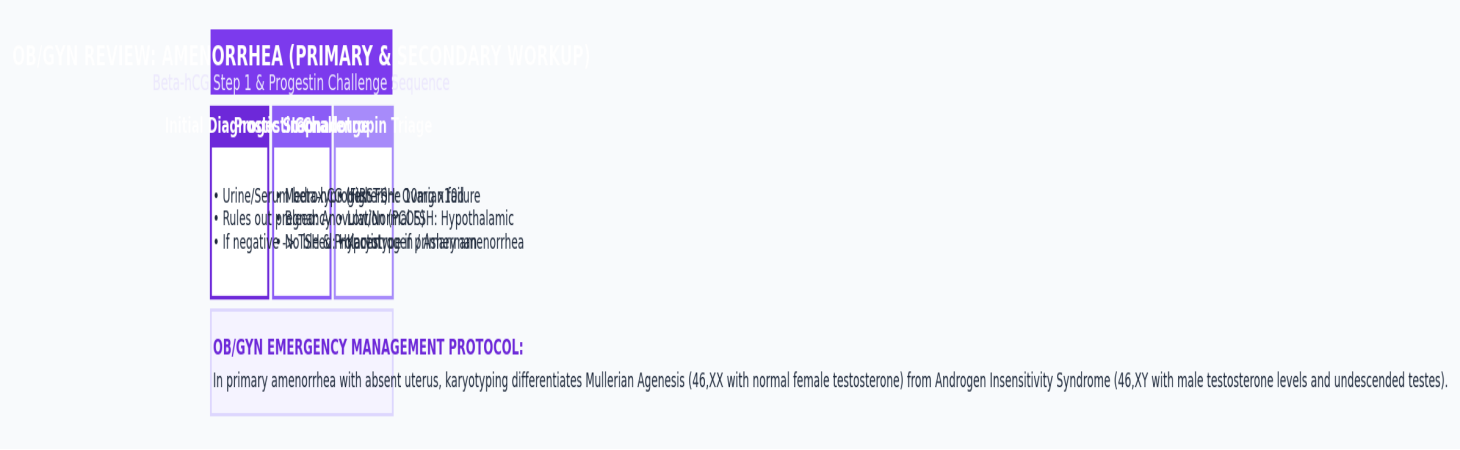
- **Presentation:** Missed periods, galactorrhea, hot flashes, hirsutism, significant weight changes or extreme athletic training (hypothalamic amenorrhea).
- **Physical Exam:** BMI, Tanner staging, thyroid exam, signs of hyperandrogenism (acne, hirsutism in PCOS), pelvic exam evaluating Mullerian structures.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

STEP 1: Urine hCG (rule out pregnancy!). If negative: Serum TSH, Prolactin, FSH, LH, and Progestin withdrawal challenge (Medroxyprogesterone 10 mg x 10d).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Target underlying etiology. Pregnancy care. Hyperprolactinemia: Cabergoline or Bromocriptine. Thyroid dysfunction treatment. PCOS: weight loss, OCPs, Metformin. Hypothalamic: caloric restoration.



HIGH-YIELD BOARD PEARL: Pregnancy is the most common cause of secondary amenorrhea; a urine or serum pregnancy test (beta-hCG) must be the first diagnostic test performed in any female of reproductive age presenting with missed menses.

Topic 2: Abnormal Uterine Bleeding (AUB: PALM-COEIN)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Uterine bleeding outside normal volume, frequency, or regularity. Etiology: PALM (Structural: Polyp, Adenomyosis, Leiomyoma, Malignancy) and COEIN (Non-structural: Coagulopathy, Ovulatory dysfunction, Endometrial, Iatrogenic, Not classified).

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Heavy menstrual bleeding, intermenstrual bleeding, postcoital bleeding, or postmenopausal bleeding (ENDOMETRIAL CANCER until proven otherwise!). Lightheadedness, syncope, pallor.
• **Physical Exam:** Pelvic speculum exam (visualize cervical lesions, foreign bodies, quantify bleeding rate), bimanual exam (uterine enlargement in fibroids, cervical motion tenderness in PID).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urine/serum pregnancy test (mandatory). CBC (anemia, thrombocytopenia), Coagulation panel (vWD). Transvaginal ultrasound (TVUS) to assess endometrial stripe thickness (>4 mm in postmenopausal women requires biopsy).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Acute heavy unstable bleeding: Two large-bore IVs, IV fluids, type and cross. High-dose IV Conjugated Estrogen (Premarin 25 mg IV Q4-6H) to promote rapid endometrial clotting, OR high-dose oral progestins. Emergency D&C; or intrauterine tamponade balloon (Bakri) if refractory. Stable: Tranexamic acid (TXA) or oral medroxyprogesterone.



HIGH-YIELD BOARD PEARL: Postmenopausal vaginal bleeding is Endometrial Cancer until proven otherwise and mandates transvaginal ultrasound and outpatient endometrial biopsy. High-dose IV estrogen is first-line medical therapy for acute hemodynamically unstable non-pregnancy AUB.

Topic 3: Ectopic Pregnancy (Rupture & Resuscitation)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Implantation of blastocyst outside the endometrial cavity, >95% occurring in the fallopian tube (ampulla most common). Risk factors: prior ectopic, history of PID/salpingitis, tubal surgery, IUD in situ, assisted reproduction.

CLINICAL PRESENTATION & PHYSICAL EXAM

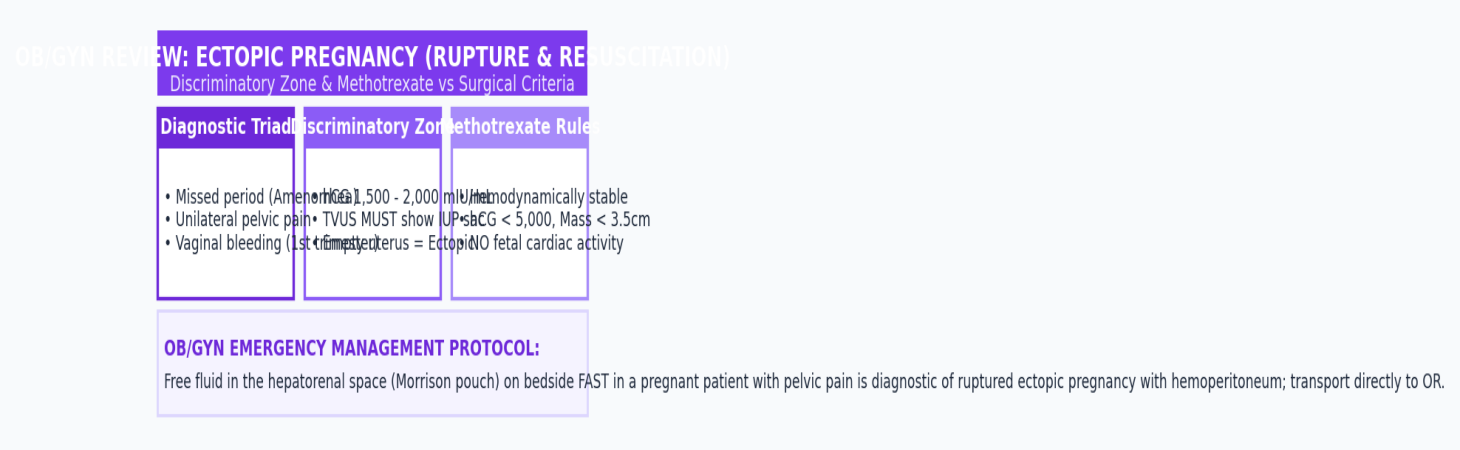
• **Presentation:** Classic triad: Amenorrhea (missed period), unilateral lower abdominal/pelvic pain, and vaginal bleeding in first trimester. Shoulder pain (Kehr sign) indicates hemoperitoneum diaphragmatic irritation.
• **Physical Exam:** Adnexal tenderness, cervical motion tenderness, palpable adnexal mass. Hypotension, tachycardia, syncope, peritoneal signs signify tubal rupture and life-threatening intra-abdominal hemorrhage.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

STAT quantitative serum beta-hCG + Transvaginal Ultrasound (TVUS). Discriminatory zone: beta-hCG 1,500-2,000 mIU/mL (an intrauterine gestational sac MUST be visible on TVUS; absence with hCG above discriminatory zone confirms ectopic). Bedside FAST exam detects free fluid in Morrison pouch and pelvis.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Ruptured / Unstable: Emergent laparoscopy/laparotomy (salpingectomy), massive transfusion protocol, STAT OB/GYN consult. Stable Unruptured: Methotrexate (50 mg/m2 IM) criteria: hCG < 5,000 mIU/mL, mass < 3.5-4 cm, no fetal cardiac activity, normal renal/liver function, patient able to comply with follow-up. Rhogam for Rh-negative mothers.



HIGH-YIELD BOARD PEARL: An empty uterus on transvaginal ultrasound with a serum beta-hCG above the discriminatory zone (1,500-2,000 mIU/mL) is an ectopic pregnancy until proven otherwise. Rh-negative mothers must receive Rh immunoglobulin (RhoGAM) to prevent alloimmunization.

Topic 4: Endometriosis

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Presence of functional endometrial glands and stroma outside the uterine cavity (ovaries [chocolate cyst / endometrioma], uterosacral ligaments, pouch of Douglas). Cyclic bleeding causes chronic inflammation and fibrosis.

CLINICAL PRESENTATION & PHYSICAL EXAM

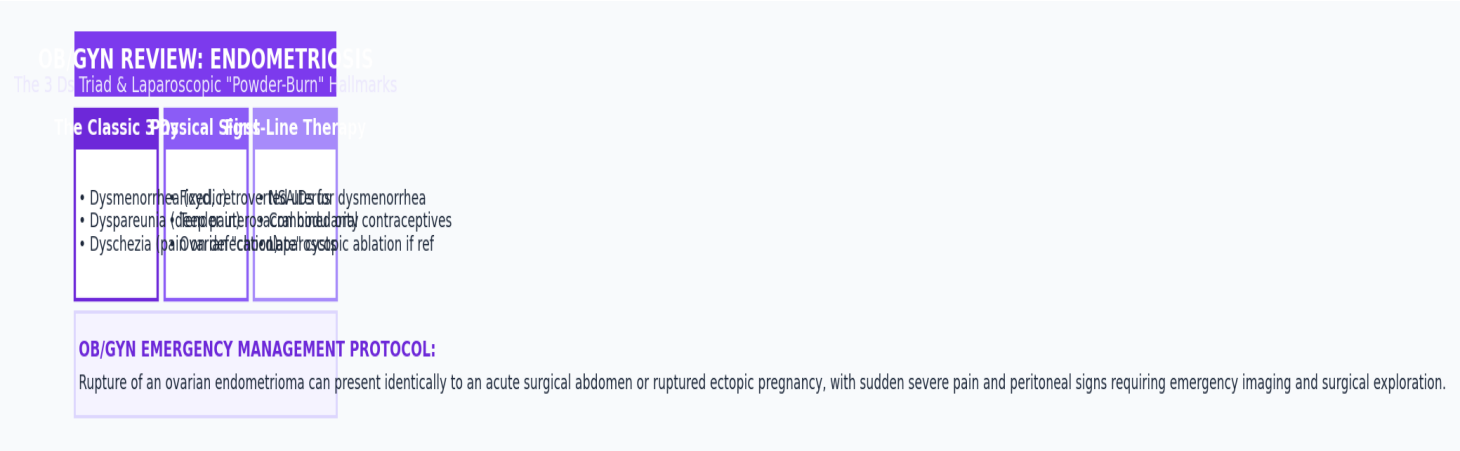
• **Presentation:** Classic "3 Ds": Dysmenorrhea (painful menses), Dyspareunia (painful intercourse), and Dyschezia (painful defecation). Infertility, chronic pelvic pain peaking prior to menstruation.
• **Physical Exam:** Fixed, retroverted uterus, tender uterosacral nodules palpable on rectovaginal exam, tender adnexal enlargement (endometrioma).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical suspicion. Pelvic ultrasound identifies ovarian endometriomas ("ground-glass" homogeneous internal echoes). Definitive diagnosis: Laparoscopy with direct visualization and biopsy ("powder-burn" lesions).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

First-line: NSAIDs for pain + Combined Oral Contraceptive Pills (OCPs) or Progestins (suppress ovulation). Second-line: GnRH agonists (Leuprolide) or GnRH antagonists (Elagolix). Surgical excision / ablation for refractory pain or infertility.



HIGH-YIELD BOARD PEARL: The classic triad of endometriosis is Dysmenorrhea, Dyspareunia, and Dyschezia ("the 3 Ds"). Definitive diagnosis requires direct visualization via laparoscopy demonstrating pathognomonic "powder-burn" lesions.

Topic 5: Fetal Distress & Intrapartum Monitoring

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Compromised fetal oxygenation during labor/pregnancy resulting in fetal hypoxemia and acidemia. Causes: uteroplacental insufficiency, umbilical cord compression, placental abruption, uterine tachysystole.

CLINICAL PRESENTATION & PHYSICAL EXAM

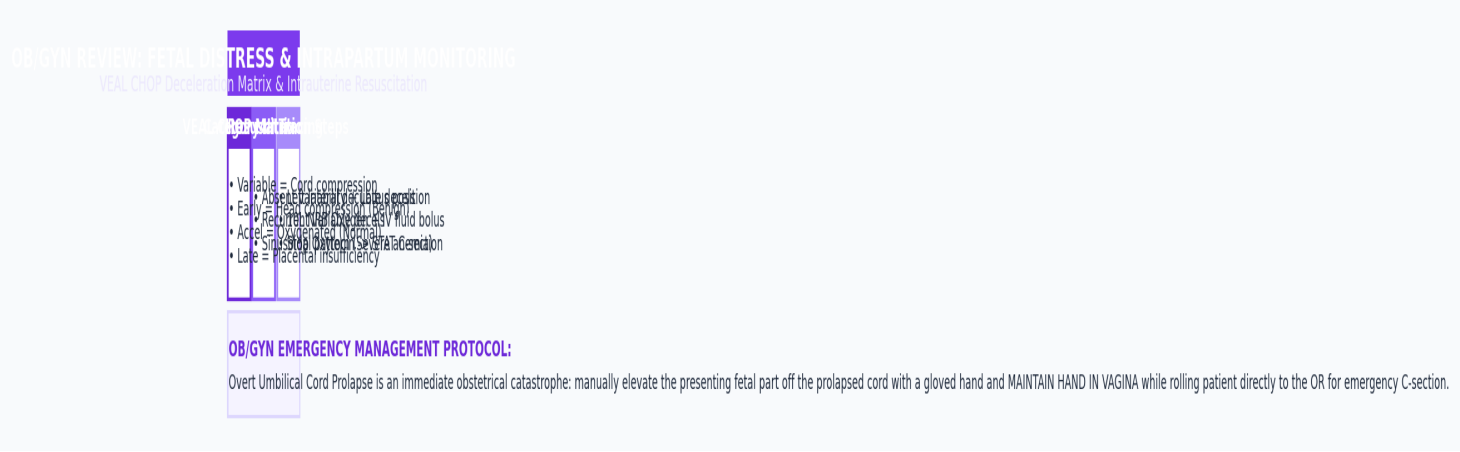
• **Presentation:** Decreased fetal movement, abnormal fetal heart rate (FHR) tracing, meconium-stained amniotic fluid.
• **Physical Exam:** Maternal vitals (hypotension, fever), uterine tenderness (abruption), umbilical cord prolapse visible on sterile speculum exam.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Electronic Fetal Monitoring (EFM): Baseline FHR (normal 110-160 bpm). Variable decelerations (cord compression - abrupt drop/rise). Early decelerations (head compression - mirrors contraction, benign). Late decelerations (uteroplacental insufficiency - begins after contraction starts, lowest point after peak; BAD!). Category III tracing (absent variability with recurrent late/variable decels, or sinusoidal pattern = severe fetal anemia).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Intrauterine Resuscitation: 1) Maternal repositioning (Left lateral decubitus); 2) Supplemental Oxygen (10 L non-rebreather); 3) IV Fluid bolus (Lactated Ringer); 4) Discontinue oxytocin/pitocin; 5) Tocolysis (Terbutaline 0.25 mg SQ) if tachysystole. STAT Emergency Cesarean Delivery if Category III tracing persists.



HIGH-YIELD BOARD PEARL: FHR Decelerations Mnemonic "VEAL CHOP": Variable = Cord compression; Early = Head compression; Acceleration = Okay; Late = Placental insufficiency. Late decelerations and sinusoidal patterns are emergency signals requiring immediate resuscitation or delivery.

Topic 6: Normal Intrauterine Pregnancy (Emergency Complications)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Physiologic changes in pregnancy: 40-50% increase in blood volume, physiologic anemia (dilutional), resting tachycardia (+10-15 bpm), relative hypotension in 2nd trimester, respiratory alkalosis (progesterone hyperventilation).

CLINICAL PRESENTATION & PHYSICAL EXAM

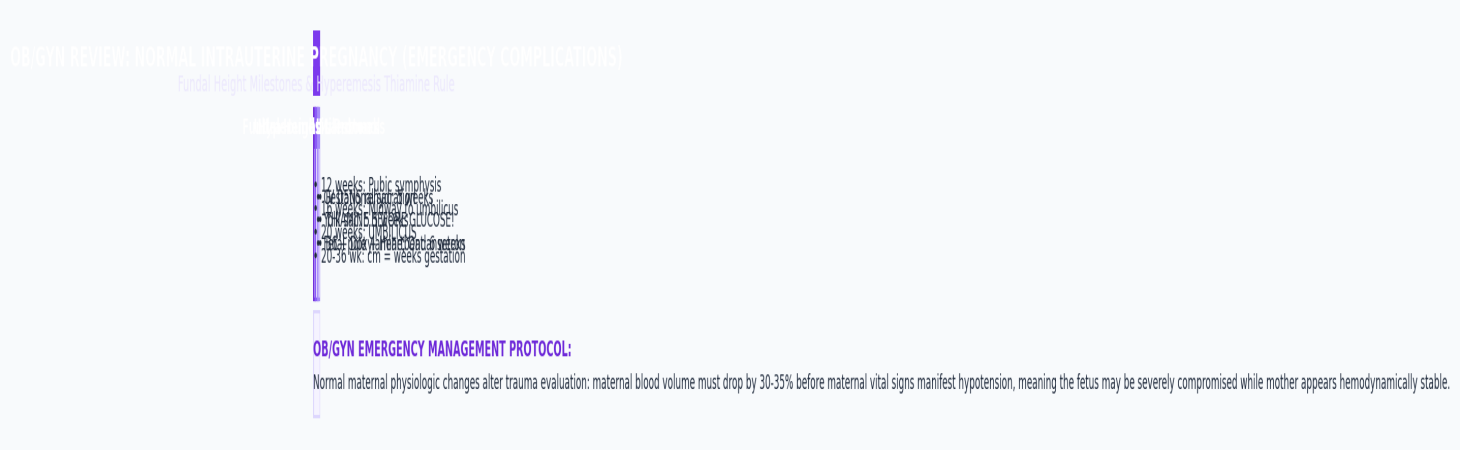
• **Presentation:** Normal early pregnancy: morning sickness, breast tenderness, fatigue. Emergencies: Hyperemesis Gravidarum (intractable vomiting, weight loss >5%, ketonuria, electrolyte derangements).
• **Physical Exam:** Uterine fundal height: 12 weeks (pubic symphysis), 20 weeks (umbilicus), 36 weeks (xiphoid process). Normal fetal heart tones audible by Doppler at 10-12 weeks.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urine hCG, qualitative/quantitative beta-hCG. Ultrasound: Gestational sac at 4.5-5 weeks, Yolk sac at 5.5 weeks, Fetal pole with cardiac activity at 6 weeks. Electrolytes, urine ketones, LFTs for hyperemesis.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Hyperemesis Gravidarum: IV rehydration with D5 Normal Saline (ADD THIAMINE 100 mg BEFORE DEXTROSE to prevent Wernicke Encephalopathy!). Antiemetics: Pyridoxine (Vit B6) + Doxylamine (first-line); Ondansetron, Promethazine, Metoclopramide.



HIGH-YIELD BOARD PEARL: Always administer IV Thiamine (Vitamin B1) prior to or concurrently with intravenous dextrose in patients with Hyperemesis Gravidarum to prevent triggering acute, irreversible Wernicke Encephalopathy.

Topic 7: Mastitis & Breast Abscess

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Infection of lactating breast parenchyma due to milk stasis and cracked nipples allowing bacterial entry. Pathogen: Staphylococcus aureus (most common, including MRSA). If untreated, progresses to breast abscess.

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Lactating female with sudden onset of unilateral breast pain, localized erythema, swelling, warmth, accompanied by high fever, chills, and myalgias.
• **Physical Exam:** Focal wedge-shaped erythema, induration, and tenderness. A localized fluctuant, exquisitely tender, palpable mass indicates Breast Abscess.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical diagnosis. Breast Ultrasound is the imaging modality of choice to identify or rule out drainable abscess.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Lactational Mastitis: Dicloxacillin 500 mg QID x 10-14d OR Cephalexin 500 mg QID. MRSA risk: Clindamycin or TMP-SMX (avoid Bactrim if infant < 2 months). CRUCIAL: MOTHER MUST CONTINUE BREASTFEEDING or pumping to relieve milk stasis (milk is safe for infant). Breast Abscess: Needle aspiration (ultrasound-guided) or surgical I&D; + systemic antibiotics.



HIGH-YIELD BOARD PEARL: Mothers with mastitis must be strongly encouraged to CONTINUE BREASTFEEDING or pumping from the affected breast; milk stasis perpetuates the infection, and continued nursing does NOT harm the infant.

Topic 8: Ovarian Cysts & Rupture

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Functional ovarian cysts: Follicular (failure to rupture) or Corpus Luteum (failure to regress, vascular, prone to hemorrhage). Complications: Cyst rupture with hemoperitoneum, or triggering Ovarian Torsion.

CLINICAL PRESENTATION & PHYSICAL EXAM

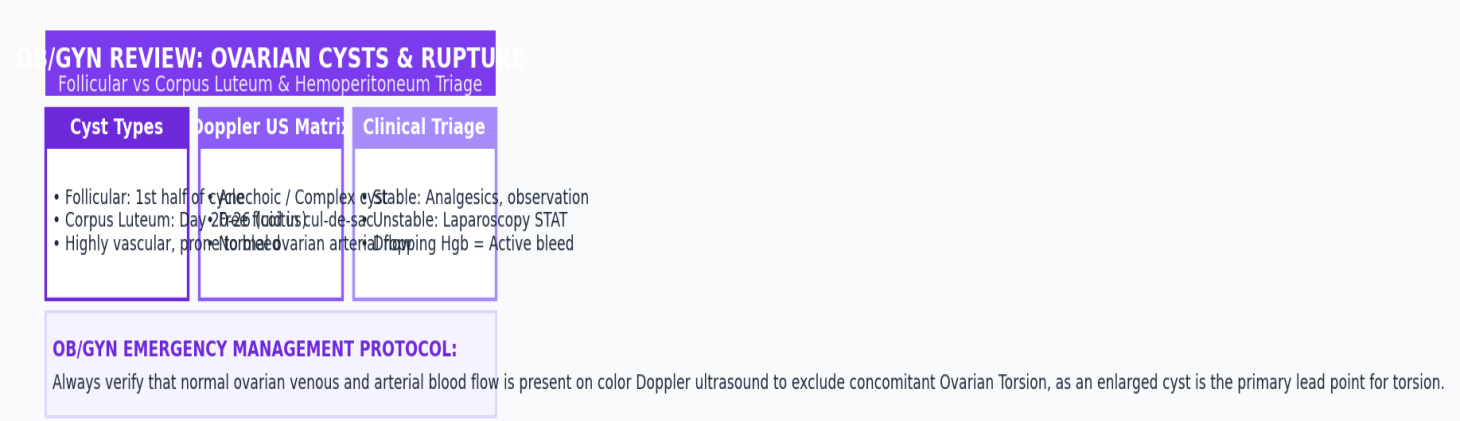
• **Presentation:** Sudden onset of unilateral lower quadrant pelvic pain, often following sexual intercourse or strenuous physical activity. Mild vaginal spotting.
• **Physical Exam:** Unilateral lower abdominal and adnexal tenderness. Guarding/rigidity and peritoneal signs suggest extensive hemoperitoneum.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Urine pregnancy test (mandatory). Transvaginal Pelvic Ultrasound with Doppler: reveals adnexal cystic structure and free pelvic fluid in pouch of Douglas. CBC (monitor hematocrit).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Uncomplicated rupture (physiologic fluid, stable vitals, mild pain): Outpatient supportive care with oral analgesics and rest. Complicated rupture (ongoing hemorrhage, hemodynamic instability, drop in hematocrit): Hospital admission, IV fluid resuscitation, type & cross, emergent laparoscopy for surgical hemostasis.



HIGH-YIELD BOARD PEARL: Ruptured corpus luteum cysts typically occur in the second half of the menstrual cycle (day 20-26) often triggered by coitus. Hemodynamically stable patients without ongoing bleeding can be managed conservatively with close observation.

Topic 9: Pelvic Inflammatory Disease (PID) & TOA

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Ascending polymicrobial infection of the female upper genital tract (endometrium, fallopian tubes, ovaries, peritoneum). Pathogens: Chlamydia trachomatis, Neisseria gonorrhoeae, anaerobes, enteric gram-negative rods, Mycoplasma genitalium.

CLINICAL PRESENTATION & PHYSICAL EXAM

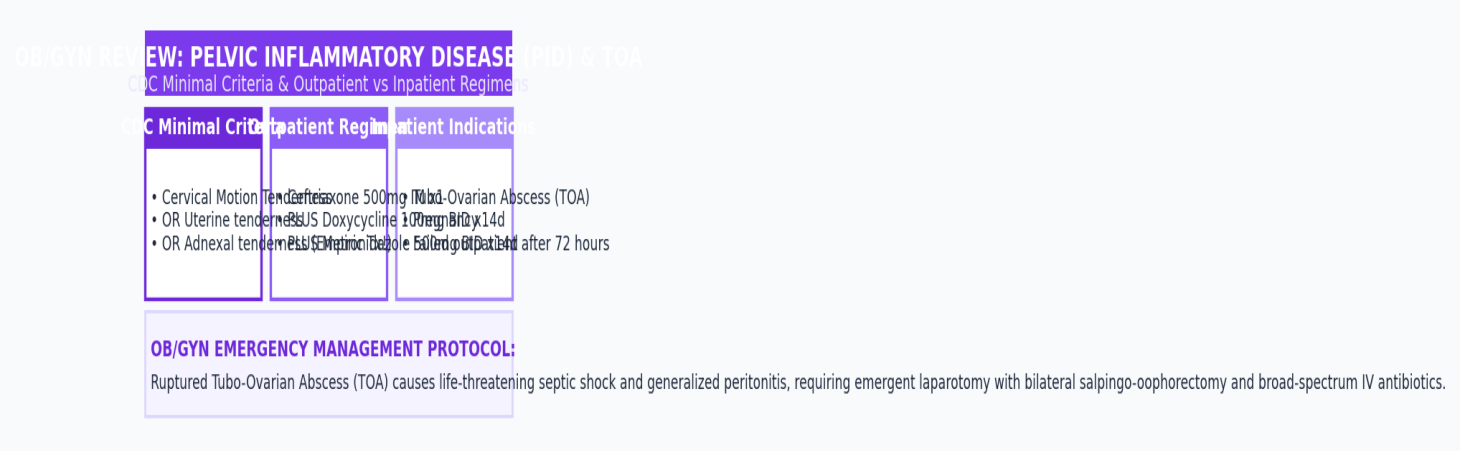
• **Presentation:** Lower abdominal and pelvic pain, abnormal vaginal discharge, dyspareunia, postcoital bleeding, fever. Right upper quadrant pain (Fitz-Hugh-Curtis syndrome: perihepatitis "violin-string" adhesions).
• **Physical Exam:** CDC Minimal Diagnostic Criteria (at least ONE in a sexually active female with pelvic pain): 1) Cervical Motion Tenderness ("Chandelier sign"); 2) Uterine tenderness; OR 3) Adnexal tenderness. Mucopurulent cervical discharge.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Pregnancy test (rule out ectopic). Wet mount (abundant WBCs), NAAT for GC/Chlamydia. Transvaginal Ultrasound: thick fluid-filled tubes, or complex multilocular adnexal mass with debris = Tubo-Ovarian Abscess (TOA).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Outpatient (mild/moderate): Ceftriaxone 500 mg IM single dose + Doxycycline 100 mg PO BID x 14 days + Metronidazole 500 mg PO BID x 14 days. Inpatient Indications: TOA, pregnancy, cannot exclude appendicitis/surgical abdomen, severe illness/vomiting, failed outpatient therapy. Inpatient IV Regimen: Cefotetan 2g IV Q12H (or Cefoxitin) + Doxycycline 100 mg PO/IV Q12H; OR Ampicillin-sulbactam + Doxycycline.



HIGH-YIELD BOARD PEARL: According to CDC guidelines, empiric treatment for PID should be initiated in any sexually active young female presenting with pelvic pain if Cervical Motion Tenderness ("Chandelier sign"), uterine, or adnexal tenderness is present on bimanual exam.

Topic 10: Pelvic Pain & Dysmenorrhea

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Primary Dysmenorrhea: excess uterine prostaglandin (PGF2a) production causing painful myometrial contractions and uterine ischemia. Secondary Dysmenorrhea: underlying pelvic pathology (endometriosis, adenomyosis, fibroids, PID, pelvic congestion).

CLINICAL PRESENTATION & PHYSICAL EXAM

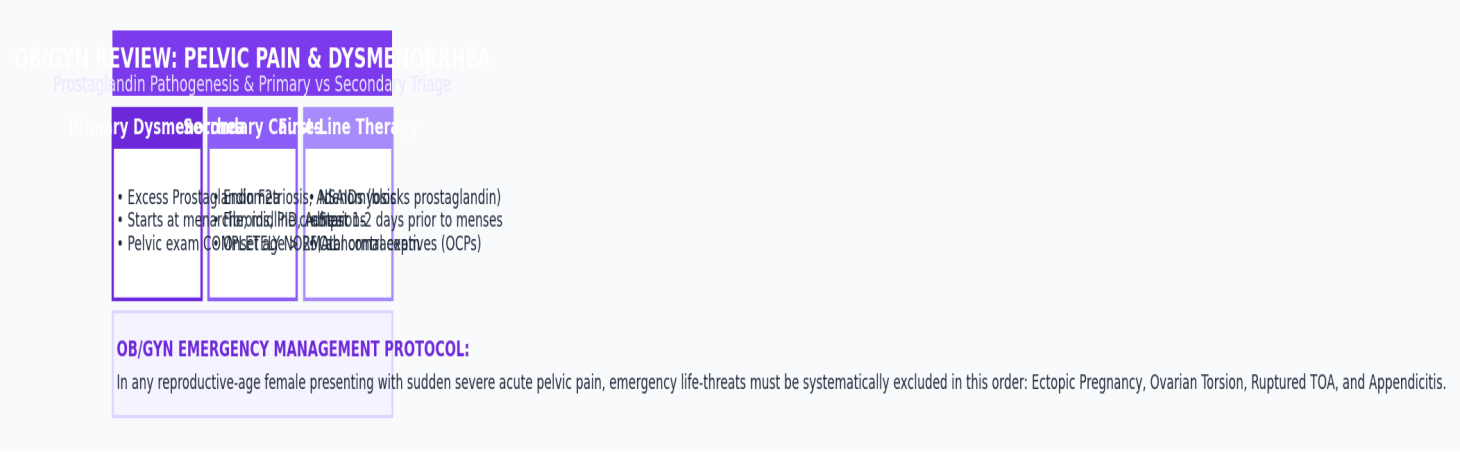
• **Presentation:** Primary: cramping lower midline abdominal/pelvic pain starting 1-2 days before or with onset of menses, radiating to lower back/thighs, lasting 24-72 hours. Associated nausea, diarrhea, fatigue. Secondary: pain non-midline, starts later in life, persists beyond menses.
• **Physical Exam:** Primary dysmenorrhea has a COMPLETELY NORMAL pelvic examination. Abnormalities (fixed uterus, pelvic mass, tenderness outside menses) indicate secondary causes.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Pregnancy test (negative). Clinical diagnosis for primary. Transvaginal pelvic ultrasound to evaluate for secondary anatomical causes if pelvic exam abnormal or unresponsive to therapy.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

First-line: NSAIDs (Ibuprofen 600-800 mg TID or Naproxen 500 mg BID; start 1-2 days before expected menses to block prostaglandin synthesis). Second-line: Combined oral contraceptives (OCPs) or levonorgestrel IUD (suppresses ovulation and thins endometrium). Local heat therapy.



HIGH-YIELD BOARD PEARL: Primary dysmenorrhea is mediated by uterine Prostaglandin F2-alpha (PGF2a) and features a completely normal physical and pelvic examination. NSAIDs are first-line therapy because they directly inhibit prostaglandin production.

Topic 11: Placental Abruption (Abruptio Placentae)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Premature detachment of a normally implanted placenta from the uterine wall prior to delivery. Risk factors: MATERNAL HYPERTENSION / PREECLAMPSIA (most common), cocaine use, abdominal trauma, smoking, rapid uterine decompression.

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Classic triad: Painful 3rd-trimester vaginal bleeding (80%), severe continuous abdominal/back pain, and hypertonic, frequent uterine contractions. (Bleeding may be 20% concealed retroplacental!).
• **Physical Exam:** Uterine tenderness, high resting uterine baseline tone (woody, rigid uterus). Fetal distress (late decelerations, bradycardia). Maternal shock out of proportion to visible vaginal bleeding.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Clinical diagnosis. Ultrasound sensitivity is POOR (<50%; retroplacental clot cannot exclude abruption). Coagulation panel (STAT Fibrinogen, PT/INR, D-dimer, Platelets) to detect Disseminated Intravascular Coagulation (DIC). Fetal cardiotocography monitoring.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Two large-bore IVs, crystalloid resuscitation, massive transfusion protocol if shock. Continuous fetal monitoring. Emergent Cesarean delivery for maternal instability, coagulopathy, or persistent fetal distress. Vaginal delivery considered only if stable and fetal demise has occurred. Treat DIC with FFP, Cryoprecipitate (if fibrinogen < 150 mg/dL), and platelets.



HIGH-YIELD BOARD PEARL: Placental abruption presents with PAINFUL vaginal bleeding and a rigid, hypertonic ("woody") uterus, whereas placenta previa presents with PAINLESS bleeding. Ultrasound is unreliable (frequently misses retroplacental hematomas); normal ultrasound NEVER rules out abruption.

Topic 12: Placenta Previa

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Placental tissue overlying or adjacent to the internal cervical os. Complete: covers os completely; Partial; Marginal: within 2 cm of os. Risk factors: prior Cesarean delivery, prior previa, multiparity, advanced maternal age, multiple gestation.

CLINICAL PRESENTATION & PHYSICAL EXAM

• **Presentation:** Sudden onset of painless, bright red vaginal bleeding in the second or third trimester (classically around 30-34 weeks). Absence of abdominal pain or uterine tenderness.
• **Physical Exam:** Soft, nontender, relaxed uterus. DIGITAL PELVIC EXAMINATION IS STRICTLY CONTRAINDICATED (can disrupt placental attachment and provoke catastrophic, exsanguinating maternal hemorrhage!).

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Transabdominal Ultrasound (initial screening) followed by Transvaginal Ultrasound (gold standard for placental localization; safe when probe angle is controlled).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Strict Pelvic Rest (no intercourse, no digital exams). Hemodynamically stable & preterm (<36 weeks): Inpatient admission, observation, bed rest, betamethasone for fetal lung maturity. Unstable maternal hemorrhage or fetal distress: Emergent Cesarean Section regardless of gestational age. Scheduled C-section at 36-37 weeks for persistent complete previa.



HIGH-YIELD BOARD PEARL: DIGITAL VAGINAL EXAMINATION IS STRICTLY CONTRAINDICATED in third-trimester vaginal bleeding until Placenta Previa is excluded by ultrasound; touching the placenta can trigger catastrophic fatal hemorrhage.

Topic 13: Prelabor Rupture of Membranes (PROM & PPROM)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Rupture of amniotic sac prior to the onset of labor. Term PROM: >= 37 weeks. Preterm PROM (PPROM): < 37 weeks. Major complications: Intra-amniotic infection (Chorioamnionitis), cord prolapse, abruptio placentae, preterm delivery.

CLINICAL PRESENTATION & PHYSICAL EXAM

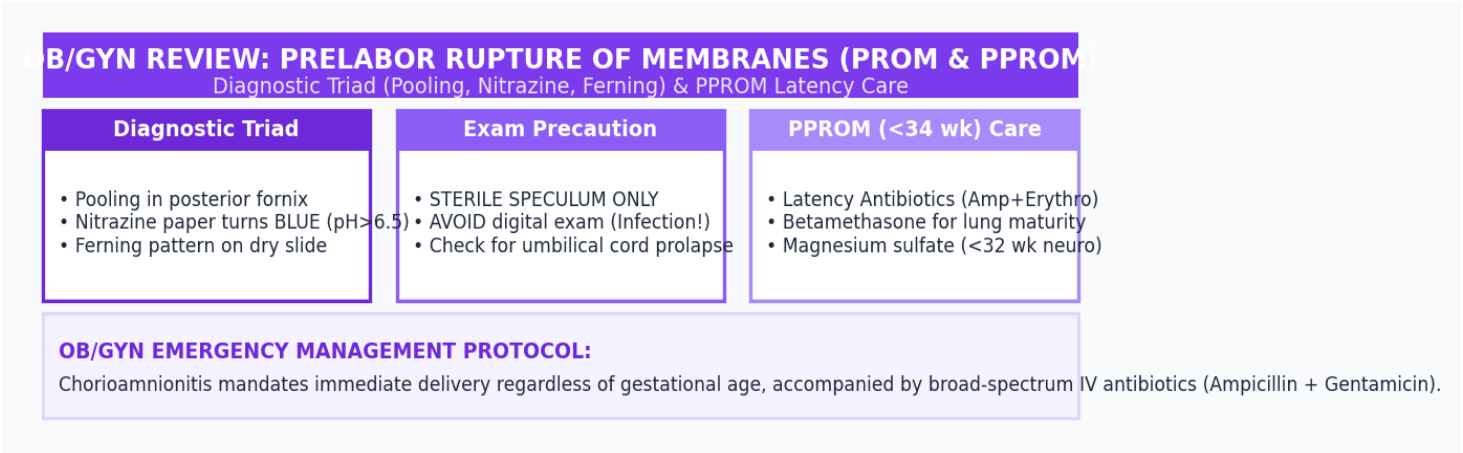
• **Presentation:** Sudden gush of clear or pale-yellow fluid from the vagina, or persistent continuous watery leakage.
• **Physical Exam:** STERILE SPECULUM EXAMINATION ONLY (avoid digital exams to prevent ascending infection). Observe for fluid pooling in posterior fornix.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Diagnostic Triad: 1) Ferning: arborization / fern pattern under light microscopy as amniotic fluid dries; 2) Nitrazine test: paper turns BLUE (pH > 6.5; amniotic fluid is alkaline); 3) Pooling: clear fluid in posterior vaginal fornix. Commercial biomarker kits (AmniSure / Actim PROM). Ultrasound shows oligohydramnios.

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Term PROM (>= 37 wk): Delivery (induction of labor with Oxytocin). PPROM (< 34 wk): Expectant latency management: 1) Latency Antibiotics (Ampicillin + Erythromycin / Azithromycin x 7 days); 2) Antenatal Corticosteroids (Betamethasone 12 mg IM Q24H x 2 doses) for lung maturity; 3) Magnesium sulfate for neuroprotection if < 32 weeks. STAT Delivery at any gestational age if Chorioamnionitis develops (fever, uterine tenderness, maternal/fetal tachycardia).



HIGH-YIELD BOARD PEARL: Diagnosis of rupture of membranes relies on the clinical triad of fluid Pooling in posterior fornix, positive Nitrazine blue test (alkaline pH > 6.5), and microscopic Ferning. In PPROM (<34 weeks), latency antibiotics and betamethasone significantly improve neonatal survival.

Topic 14: Spontaneous Abortion (First-Trimester Triage)

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Pregnancy loss before 20 weeks gestation. Most common etiology: Chromosomal abnormalities (>50%, predominantly autosomal trisomies).

CLINICAL PRESENTATION & PHYSICAL EXAM

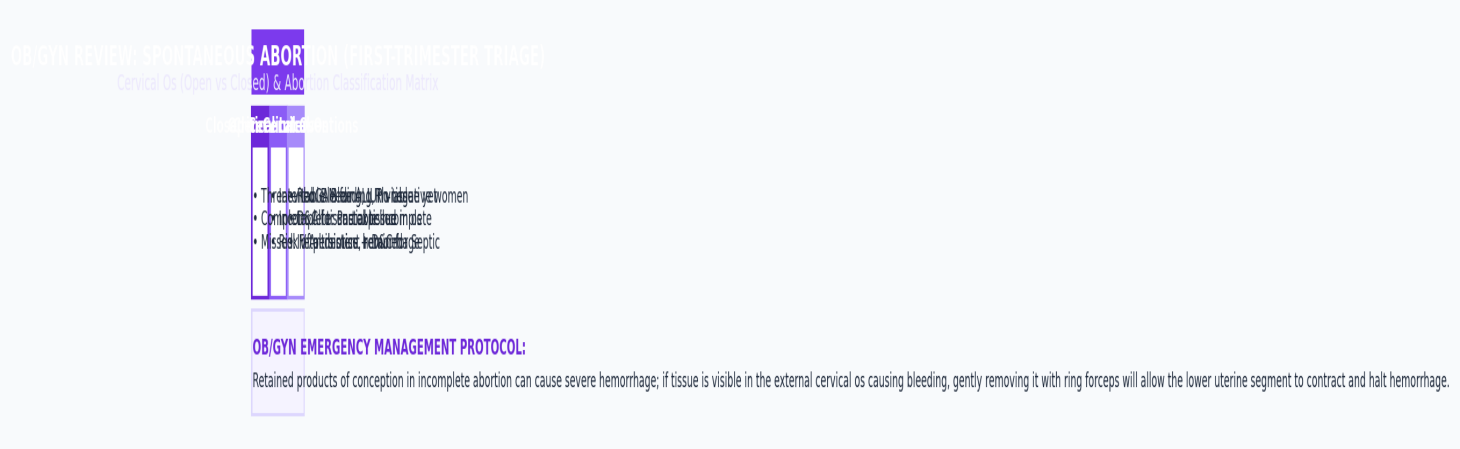
• **Presentation:** Vaginal bleeding, cramping lower midline abdominal pain, passage of tissue/clots in a pregnant patient < 20 weeks.
• **Physical Exam:** Speculum exam to evaluate the CERVICAL OS (open vs closed) and identify visible tissue in os or vaginal vault.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Classification Matrix: Threatened (Bleeding, Os CLOSED, IUP viable); Inevitable (Bleeding, Os OPEN, no tissue passed yet); Incomplete (Bleeding, Os OPEN, some tissue passed); Complete (Bleeding subsided, Os CLOSED, all tissue passed, empty uterus on US); Missed (Fetal demise, Os CLOSED, no tissue passed, no bleeding); Septic (Infection, foul discharge, fever, cervical motion tenderness).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Threatened: Pelvic rest, expectant observation, repeat hCG in 48h. Incomplete / Inevitable: Medical (Misoprostol) or Surgical (Dilation & Curettage [D&C;] or Manual Vacuum Aspiration) if heavy bleeding. Septic: Broad-spectrum IV antibiotics (Ampicillin/Sulbactam + Gentamicin) + STAT uterine evacuation. ALL Rh-negative mothers must receive RhoGAM (50 mcg in 1st trimester; 300 mcg if > 12 weeks).



HIGH-YIELD BOARD PEARL: The cervical os is CLOSED in Threatened, Complete, and Missed abortions; the cervical os is OPEN in Inevitable and Incomplete abortions. All Rh-negative unsensitized pregnant women with vaginal bleeding must receive RhoGAM.

Topic 15: Vaginitis & Vulvovaginal Infections

EMERGENCY MEDICINE CLINICAL PRACTICE & BOARD SPECIFICATION

PATHOPHYSIOLOGY & ETIOLOGY

Disruption of normal lactobacillus-dominant vaginal flora. 1) Bacterial Vaginosis (BV): Gardnerella vaginalis overgrowth; 2) Trichomoniasis: Trichomonas vaginalis (flagellated protozoan, STI); 3) Vulvovaginal Candidiasis: Candida albicans overgrowth.

CLINICAL PRESENTATION & PHYSICAL EXAM

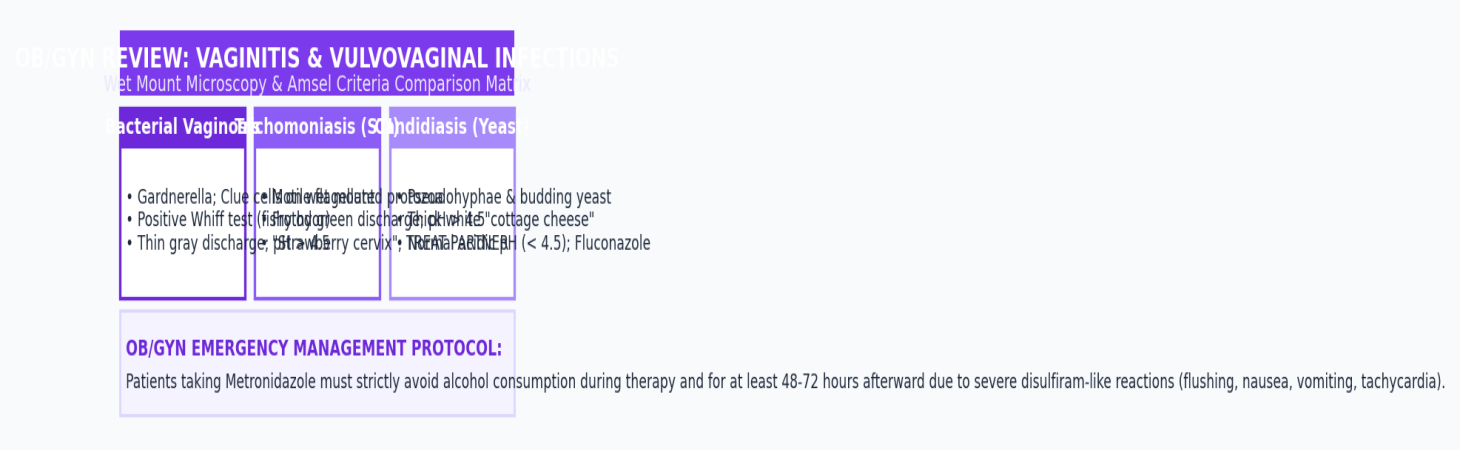
• **Presentation:** BV: Thin, gray-white homogeneous discharge with fishy odor. Trichomoniasis: Copious, frothy, yellow-green malodorous discharge, severe pruritus, dysuria. Candidiasis: Thick, white, curdy "cottage cheese" discharge, intense vulvar pruritus, burning.
• **Physical Exam:** BV: No vulvar inflammation. Trich: Strawberry cervix (punctate cervical hemorrhages). Candidiasis: Marked vulvar and vaginal erythema, edema, satellite excoriations.

DIAGNOSTIC STUDIES & LABORATORY TRIAGE

Wet mount microscopy & Whiff test (KOH). BV (Amsel Criteria, 3 of 4): 1) Clue cells (>20%); 2) Positive Whiff test (fishy amine odor with 10% KOH); 3) Vaginal pH > 4.5; 4) Homogeneous discharge. Trich: Motile flagellated trichomonads, pH > 4.5. Candidiasis: Pseudohyphae and budding yeast on KOH prep, normal acidic pH (< 4.5).

MANAGEMENT & EMERGENCY RESUSCITATION PROTOCOLS

Bacterial Vaginosis: Metronidazole 500 mg PO BID x 7 days OR Metronidazole 0.75% vaginal gel x 5 days (no partner treatment needed). Trichomoniasis: Metronidazole 500 mg PO BID x 7 days (or 2g single dose); TREAT ALL SEXUAL PARTNERS! Candidiasis: Oral Fluconazole 150 mg single dose OR topical Clotrimazole/Miconazole.



HIGH-YIELD BOARD PEARL: Clue cells and a positive Whiff test indicate Bacterial Vaginosis (treat with Metronidazole; partner treatment NOT needed). Trichomoniasis shows motile flagellated protozoa and strawberry cervix (treat with Metronidazole; PARTNER TREATMENT MANDATORY).